# A Glanceable Visual Notation for Jazz Performance Routines

One round-by-round strip for drummers, rhythm section and front line

Ben

Red Jackets Jazzband

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**Abstract.** *Background.* Jazz bands agree their routines in rehearsal: the order of choruses, who solos, where the band drops out or stops. Existing music encodings represent the order of sections but not these roles. Corpora of structural annotations record roles only after the fact, for retrieval research. *Aim.* We design one shared visual notation of the routine that every member of the band can read at a glance while performing: drummers, the rest of the rhythm section and the front line, including players who do not read the lead sheet, such as drummers and many vocalists. *Method.* We extend an existing strip of arrows, one per round, used in a traditional-jazz band's online songbook. Three kinds of information get a fixed place and a distinct mark: drop-outs, drawn as minus lines paired with the existing plus lines; solo order, drawn as a numbered sequence with lengths; and feel, split into groove, texture and dynamics. Groove is drawn only where it changes; texture and dynamics apply to one round. The design draws on research on preattentive visual features. We coded the band's 16 existing routines (83 rounds) against categories derived from the extensions. *Results.* In 13 rounds (16%), at least one category appeared only as an unstructured remark. Four of these remarks could not be classified without the band's input. Of 19 solo rounds, 14 named no soloist. *Conclusion.* The notation covers the corpus, and the coding exposed ambiguity in the band's existing wording. Whether the strip is faster to read than written remarks is untested. We specify a within-subjects study to test it.

**Keywords:** jazz, performance routine, head arrangement, music notation, information visualisation, rehearsal

## 1 Introduction

A jazz performance has two layers of structure. The *form* belongs to the composition: a 12-bar blues, a 32-bar AABA song, a multi-strain march. The *routine* belongs to a particular band's performance: the intro, the number of choruses, who sings, who solos and for how long, where the band stops, how the tune ends. Berliner (1994, chs. 11–12) describes how such arrangements are decided in rehearsal and filled in through shared conventions.

An unwritten routine depends on every player remembering it. That assumption is weakest where rehearsal is most needed: with a new member, a stand-in, or a tune not played for months. A written routine helps only if it can be read in the moment between two phrases, while the eyes are also needed for the music. Sentences of remarks under a lead sheet are a poor fit for that task, because each must be read word by word.

The routine also matters most to the players least served by the lead sheet. A drummer has no melody or chords to follow, and in our experience many vocalists sing from the lyrics and do not read the chart. Yet both need the routine: the drummer to set up breaks, stop time and the change of groove, the vocalist to know which choruses are theirs. A lead sheet gives them nothing to look at. A routine strip needs no ability to read notes or chord symbols, so it can be the one written part they actually use. Drummers already write their own form charts for this reason (Section 2.3).

Our aim is therefore a *universal* notation of the routine: one strip that the drummer, the rest of the rhythm section (bass or sousaphone, banjo, piano) and the front line (trumpet, clarinet, trombone, voice) all read, instead of a lead sheet for some and a private form chart for others. Each role asks different questions of the routine, but the answers should sit on the same page, in the same places, for everyone.

We address this with a visual notation of the routine. It extends a strip of arrows, one per round, that the songbook of the Red Jackets Jazzband (a traditional-jazz band) draws under each tune's chord grid. The contributions are:

1. a review of how routines are represented in music encodings and structure-annotation corpora, and of the perceptual basis for reading a display at a glance (Section 2);
2. three extensions to the strip (drop-outs, solo order and a three-axis model of feel), each with a fixed position, a distinct mark and an explicit scope (Section 3);
3. an analysis of where each role in the band finds its answers on the strip (Section 3.5);
4. a coding of the band's 16 routines against the extensions (Section 4), and a protocol for testing whether the strip is read faster than remarks (Section 5).

The paper concerns the visual form only. How a routine is entered or stored is out of scope.

## 2 Related work

### 2.1 Notation and encoding standards

Conventional notation marks the order of sections inside the music, with rehearsal letters, repeat signs, first and second endings, and *D.S.*, coda and *Fine* jumps. These symbols say where to go next. They do not say who plays.

Digital encodings carry the same information. The ABC standard lets a tune header list the play order of lettered parts, with repeat counts and grouping (Walshaw, 2011). MusicXML 4.0 encodes repeats, endings and jumps, and its `time-only` attribute restricts an element to particular passes through a repeated section (W3C Music Notation Community Group, 2021). MEI's `<expansion>` element gives the performed order of sections as an ordered list, and a piece may carry several expansions for different readings of its repeats (Music Encoding Initiative, n.d.). All three represent *order*. None has a concept of a soloist, an instrument sitting out, or a change of feel.

### 2.2 Structure-annotation corpora

Music information retrieval has produced corpora that annotate the structure of recordings. SALAMI (Smith et al., 2011) holds over 2,400 annotations of nearly 1,400 recordings, with musical similarity, function and instrumentation in separate layers. JAAH (Eremenko et al., 2018) annotates meter, structure and chords for 113 jazz tracks. JSD (Balke et al., 2022) segments 340 jazz recordings into over 3,000 segments; each segment records the predominant instrument, usually the soloist, and the accompanying instruments. The Weimar Jazz Database (Pfleiderer et al., 2017) annotates transcribed solos with chorus and form structure and with per-solo metadata, including a rhythm feel, defined as the basic groove organisation of the solo.

These corpora show that roles and feel can be annotated systematically. JSD's pairing of soloist and accompaniment and the database's per-solo rhythm feel correspond directly to two of our extensions. But the annotations are *descriptive*: made after the fact, time-stamped to a recording, and designed for computational analysis. A routine is *prescriptive* and must be read during performance.

### 2.3 Practitioners' form charts

Drummers have their own, informal answer to the same problem: a one-page chart of the song's whole form with only a few bars of drum notation. It outlines the sections with their length in bars ("Verse – 8") plus short cues such as a groove, a hit or an energy level (e.g., Austin Stone, 2017; Bergamini, 2021), which Bergamini (2021) calls the most important part; Ficalora's Drum Chart Builder app likewise maps song form, grooves and hits (Ficalora, n.d.), and the trade press offers similar guidance (Drum! Magazine, n.d.).

Beyond drums, the Nashville number system gives the whole band a compact, section-by-section chart for players who do not read notation (Williams, 2012). A blog post by Hein (2012) draws songs as colour-coded blocks along a bar timeline, a visual precedent for the strip's part boxes that records only section order, length and type. A web template by Scott (n.d.) is a contrast: a planning aid for the writer that says nothing about who plays in performance.

These charts confirm the need this paper starts from: musicians already reduce a song to its form plus a few cues. The strip differs in being shared by the whole band, where drum charts are personal, and in leaving bar counts to the tune's known form.

### 2.4 Reading a display at a glance

Feature-integration theory holds that simple visual features are registered early and in parallel across the visual field, while combinations of features require serial, focused attention (Treisman & Gelade, 1980). Healey & Enns (2012) survey how visualisation design uses such preattentive features to make a target found without search. For a routine, this argues for answers marked by a distinct shape in a fixed position rather than embedded in prose, which must be read serially.

The reader is also a performer. Eye-movement research on music reading is much less developed than research on reading text (Madell & Hébert, 2008), and its methods vary widely between studies (Puurtinen, 2018). We are not aware of any study of how musicians read arrangement or routine information during performance. Table 1 summarises what each representation records.

## 3 Design

### 3.1 Requirements

The target task is a player answering *"what do I do this round?"* in one glance, between phrases. Five requirements follow.

**R1 Fixed place, distinct mark.** Each kind of answer occupies the same position in every round and has a mark unlike the others, so it can be found without reading (Treisman & Gelade, 1980; Healey & Enns, 2012).

**R2 One place per fact.** A fact appears in exactly one position, so the player never has to check two.

**R3 Familiar marks.** Reuse signs musicians already read (plus and minus, numbered order, repeat counts, the widening hairpin of a crescendo) rather than invented icons.

**R4 Quiet by default.** A round with nothing unusual adds no marks, so unusual rounds stand out.

**R5 Greyscale.** Meaning is carried by position, typography and signs, not colour, so a printed copy loses nothing.

**R6 Role-neutral.** Nothing requires reading pitches, chord symbols or instrument-specific notation, so every player, whatever their instrument or reading habits, can use the same strip.

### 3.2 Baseline

The existing strip runs left to right under the chord grid, one chevron-shaped arrow per round (Figure 1). Inside each arrow are the round number, one box per part played (A, B, Intro) and a repeat count ("4x"); an ellipsis box marks a round that lasts until a cue. Rounds playing the same part share a shade of grey. Under the arrow come the *lead* (who carries the round), underlined;

**Table 1.** What existing representations record about a routine.

| Source | Section order | Who plays | Solo order | Feel | Purpose |
|---|---|---|---|---|---|
| Conventional notation | in the music | no | no | style or tempo indication | performance |
| ABC, MusicXML | yes | no | no | no | engraving, exchange |
| MEI | yes, with variants | no | no | no | scholarly editions |
| SALAMI | yes | instrumentation layer | no | no | retrieval research |
| JSD | yes | soloist and accompaniment | implicit in segment order | no | retrieval research |
| Weimar Jazz Database | per solo | soloist | per solo | rhythm feel per solo | jazz research |
| Drummers' form charts | outline with bar counts | short per-section cues | no | groove and energy cues | one drummer's own reading |
| This paper | one arrow per round | lead, joins, drop-outs | numbered sequence | groove, texture, dynamics | reading during performance |

**Table 2.** Three questions, three regions of the arrow.

| Question | Region | Marks |
|---|---|---|
| **Who** plays, and who does not | under the arrow | lead; + joins; − drop-outs; numbered solo order |
| **How** the beat goes | in, above and below the arrow | arrow height for dynamics; groove label above; texture in small capitals at the foot |
| **What** else is played | remarks line | unchanged |

one "+ instrument" line for each instrument that joins; then any other remarks on a single line. The baseline already separates part, lead and joins. Everything else falls into the remarks line, the last place the eye reaches.

### 3.3 Extensions

The extensions assign three questions to three regions of the arrow (Table 2). This separation settles borderline cases. "No rhythm" answers *who*, so it is a drop-out, not a feel. A drum solo over stop-time hits answers both questions: the drummer appears in the solo order and STOP TIME at the foot.

**Drop-outs.** A "− instrument" line sits beside the "+ instrument" lines, a matched pair of signs (R3). Strikethrough was rejected because it reads as a correction. Joins and drop-outs apply to their own round only and are not cumulative, so any arrow can be read without tracing the rounds before it. This mirrors JSD's per-segment pairing of soloist and accompaniment (Balke et al., 2022), expressed as a difference from the full band.

**Solo order.** A solo round shows a numbered sequence under its lead, for example 1 sax/tbn → 2 tpt ×2 → 3 tpt. One solo is one pass through the round's parts: the jazz chorus, which the Weimar Jazz Database also annotates within each solo (Pfleiderer et al., 2017). A slash marks alternates (whichever of the two is present), and "each ×1" fixes the length but not the players. The lengths sum to the repeat count in the arrow, so the plan can be checked at a glance.

**Feel.** In band usage, "feel" covers three independent things. Each gets its own region and scope (Table 3).

The scopes differ on purpose. A groove usually holds for a whole tune and changes at most at one point. Drawing it on every arrow would violate R4, and the eye can scan the line above the strip for changes alone. Stop time, breaks and "softly behind the vocals" are events that end with their round. Half feel and double feel are relative to the groove in force, so they are treated as texture.

**Dynamics as height.** Dynamics are drawn as the height of the arrow rather than as letters such as *p* and *f*: a soft round is a lower arrow and a loud round a taller one; a crescendo widens from left to right, the shape of the hairpin players already read (R3); and normal level is the default height and adds nothing (R4). Arrows are centred on one line, so the number and part boxes stay in the same place whatever the height (R1). More sound maps to more ink, so there is no symbol to decode, and length is among the more accurately judged graphical encodings (Cleveland & McGill, 1984). Only three levels are used (soft, normal, loud), so each differs clearly from its neighbours without a scale (Figure 2).

A texture or dynamic limited to part of a round names that part ("BREAKS in the 2nd B"), using the labels of the part boxes. Labels come from a short fixed list per axis, which can be translated on the page. A label outside the list is shown as written rather than dropped.

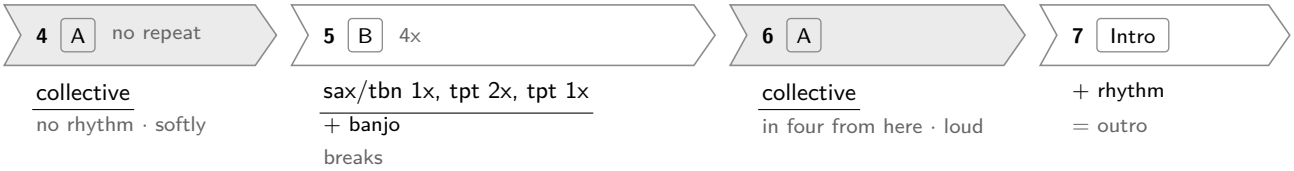



**Figure 1.** The baseline strip for rounds 4–7 of an illustrative *Dippermouth Blues* routine. Who plays and how the band plays are all in the lead and the grey remarks line, as running text. Figure 3 shows the same content in the extended strip.

**Table 3.** The three axes of feel.

| Axis | Region and mark | Scope | Labels |
|---|---|---|---|
| **Groove**: the rhythm section's beat pattern | above the arrow: ♩ and label | until changed; drawn only at a change | two-beat, four, march, second line, shuffle, straight, latin, dirge, hymn |
| **Texture**: how much of the beat is played | foot of the round, small capitals | one round | stop time, breaks, light, vamp, pedal, half feel, double feel |
| **Dynamics**: level | arrow height: lower for soft, taller for loud; widening for build | one round | soft, loud, build |

**Table 4.** What each role asks of the routine, and where it finds the answer.

| Role | Typical questions | Where on the strip |
|---|---|---|
| Drummer | Where are we in the form? Does the beat change? Stop time or breaks? How loud? Who solos next? | part boxes, groove label, texture line, arrow height, solo order |
| Rhythm section (bass, banjo, piano) | Two-beat or four? Stop time, vamp or pedal? Do I sit out? How loud? | groove label, texture line, − lines, arrow height |
| Front line (horns) | Do I lead, play or rest? When is my solo? Who joins? How loud? | lead, +/− lines, solo order, arrow height |
| Vocalist | Which rounds are mine? Who plays behind me? | lead, + lines |

### 3.4 Layout

Figure 3 shows the extended strip. Its vertical order follows the order of a player's questions: part and level, then lead, then who joins, drops out or solos, then texture, then remarks.

### 3.5 One strip for every role

The strip is meant to be universal (R6): the same drawing serves every player. Each role scans it for different questions, but every answer is in one of the fixed regions of Table 2, so no role needs its own chart (Table 4).

No region belongs to one role: the drummer and the bass player read the same groove label, and the horns and the vocalist read the same lead line. One gap remains for the drummer: the strip gives no bar counts, which drummers' own charts do (Section 2.3).

**Table 5.** Coded rounds by category ($n = 13$ rounds, 15 codes).

| Category | $n$ | Example remark | Rendered as |
|---|---|---|---|
| Texture | 7 | "Stop in the break of the 2nd B" | BREAKS in the 2nd B |
| Soloist or solo order | 4 | "sax/tbn 1x, tpt 2x, tpt 1x" | 1 sax/tbn → 2 tpt ×2 → 3 tpt |
| Drop-out | 2 | "No rhythm" | − rhythm |
| Join with timing | 1 | "bass drum joins on the 2nd line" | + bass drum from the 2nd line |
| Redundant | 1 | "With rhythm" | nothing (default) |

## 4 Corpus analysis

### 4.1 Method

The corpus is every routine in the band's songbook as of October 2026: 16 tunes with 83 rounds in total. Each round's remarks were coded against five categories defined in advance from the design: texture; soloist or solo order; drop-out; join with timing; and a remark made redundant by the defaults. A round could receive more than one code. A remark was marked *ambiguous* when its category depended on a meaning that only the band could confirm. Solo rounds were counted separately: a round counted as a solo round when its lead or remarks referred to a solo. Coding was done by a single coder, so no inter-rater agreement is reported.

### 4.2 Results

Thirteen rounds (16%) received at least one code, 15 codes in all (Table 5).

**Solo rounds.** Of 19 solo rounds, 14 (74%) gave no soloist, four named a single soloist, and one gave an order with lengths.

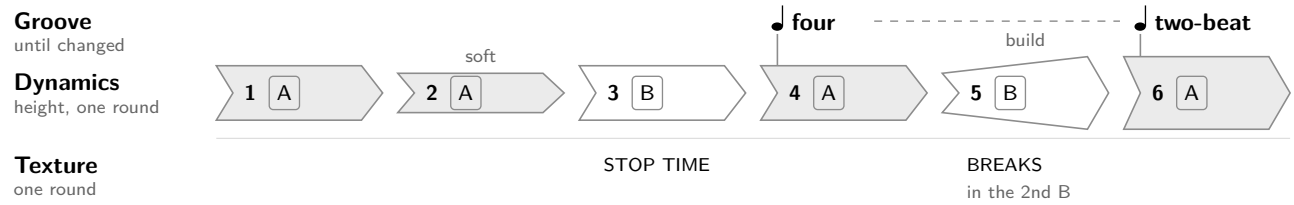



**Figure 2.** Scope rules for feel in a six-round routine. The groove is drawn only at rounds 4 and 6, where it changes, and holds in between (dashed line, an annotation). Dynamics, shown as arrow height, apply to their own round only: soft in round 2, a build in round 5, loud in round 6. So does texture (rounds 3 and 5). Round 1 has no marks, so it is played in the groove in force, with full texture, at normal level.

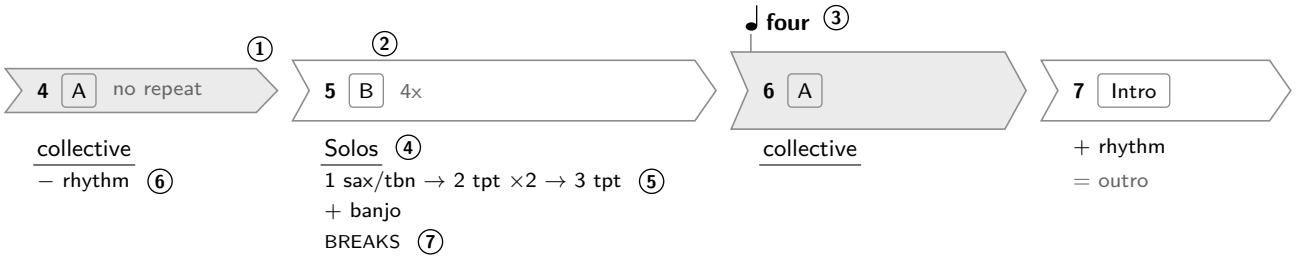



**Figure 3.** The extended strip for the same rounds as Figure 1. Callouts (annotations, not part of the notation): ① dynamics as arrow height, here soft (round 4) and loud (round 6); ② parts and repeat count; ③ groove, drawn only where it changes; ④ lead; ⑤ solo order with each soloist's length; ⑥ joins (+) and drop-outs (−), one round only; ⑦ texture in small capitals, one round only.

**Groove.** No round changed the groove, so the groove line would be empty for every routine in the corpus. Groove words do appear in the tunes' style headings (for example *second line*, *march*, *rumba*), mixed with genre words (*traditional*).

**Ambiguity.** Four of the 13 coded rounds (31%) were ambiguous:

1. "Salvation army" style: a hymn groove, or a remark about the arrangement?
2. "straight out": even eighths, or an ending without a tag?
3. "rhythm break": the band stops for the soloist's breaks, or the rhythm section takes a break?
4. "drums solo first 8 bars of the B, band back in bar 9": a drop-out of the horns, or a drum break?

In each case the remark is clear to current band members but not to a newcomer. A fixed set of labels forces the band to settle these meanings once.

## 5 Proposed evaluation

The corpus analysis shows coverage, not readability. The central claim, that the extended strip is read faster and more accurately, needs an experiment. We specify one here; it has not been run.

**Hypotheses.**

**H1.** Questions about drop-outs, solo order and feel are answered faster with the extended strip than with the same information as written remarks or as the baseline strip.

**H2.** Accuracy on those questions is higher with the extended strip.

**H3.** Questions the baseline already answers (part, lead, joins) are answered no slower with the extended strip, so the additions cost nothing for existing tasks.

**H4.** The gain from H1 and H2 holds for every role (drums, rest of the rhythm section, front line, vocals), and is at least as large for players who do not read the lead sheet as for those who do.

**Design.** A within-subjects design with three conditions: (a) remarks as plain text below the music; (b) the baseline strip (Figure 1); (c) the extended strip (Figure 3). Participants are members and stand-ins of the band, expected $n = 8$–$12$. Each is classed in advance by role and by whether they read the lead sheet while playing, to test H4; the questions asked are drawn from Table 4. Routines are assigned to conditions by Latin square, so that each routine appears in every condition across participants but only once per participant.

**Task.** Each trial shows a routine briefly (about 2 s, to be set in a pilot) and then asks one question. Examples: "In round 5, do you play?", "Who solos second?", "Is round 3 in stop time?", "Where does the beat change?", "Which round is loudest?"

**Measures and analysis.** Response time and accuracy are recorded per trial. They are analysed with mixed-effects models with crossed random effects for participants and routines (Baayen et al., 2008): linear for log response time, logistic for accuracy.

**Field measure.** Over a few rehearsals before and after adoption, count interruptions caused by questions about the routine.

**Extension.** Eye tracking during performance would show where players look for routine information, but

methods in music-reading eye tracking vary widely, and stimulus, tempo and error handling must be fixed in advance (Puurtinen, 2018).

With so few participants, only large effects are detectable. The study is exploratory.

## 6 Discussion

**Prescriptive versus descriptive structure.** The corpora in Section 2 describe what a recording did, at the resolution of seconds. A routine states what a band intends, at the resolution of a round. The strip therefore shows no times. Timing within a round is a short qualifier ("from the 2nd line"), not a position.

**Scope of the scope rules.** The distinction between a sticky groove and per-round texture and dynamics is the part of the design most likely to transfer to other repertoires. The label lists are not. They reflect one band's New Orleans repertoire, and a bebop or Latin band would need different groove labels.

**What is left out.** *Cues:* the strip shows the plan that cues steer, not who gives them; no routine in the corpus needed a cue mark. *Alternative routines:* MEI allows several performed orders per piece (Music Encoding Initiative, n.d.); the strip shows one routine per tune, and alternatives remain remarks. *Within-round positions* are not represented. *Bar counts:* drummers' own charts give each section's length in bars (Section 2.3); the strip does not, and relies on the band knowing the tune's form.

**Limitations.**

- The readability claim is grounded in perception research but untested for this display and this task (Section 5).
- The corpus is small and comes from one band.
- Coding was done by one coder, with no agreement measure.
- The designers are associated with the band whose routines were coded.
- The claim that drummers and many vocalists do not read the lead sheet comes from the band's own practice, not from a study. H4 tests whether it matters for this notation.
- Arrow height could be misread as importance or duration rather than level. The study's dynamics questions test whether it is read as intended.

## 7 Conclusion

Routine information that standard encodings do not represent can be drawn in a compact strip: who sits out, as minus lines paired with joins; who solos and for how long, as a numbered sequence; and how the band plays, split into groove, texture and dynamics, each with its own place and scope. Because none of it needs notes or chord symbols, the same strip serves the drummer, the rest of the rhythm section and the front line. Applied to one band's 16 routines, the notation covered every coded remark. It also exposed four remarks whose meaning only current members could resolve. The next steps are implementing the extended strip and running the study in Section 5.

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